



Evolution of Plume Volcanism at Atla Regio, Venus

P. J. Mason¹ , A. Klidas² , D. Cirium³ , R. C. Ghail⁴ , and S. Lea-Wurzbach¹

Key Points:

- A new regional chronostratigraphic framework is proposed for the evolution of Atla Regio
- Volcanism began along Dali Chasma, then plume volcanism began at Ozza Mons and finally Maat Mons became the site of active volcanism
- The pattern of shield volcanoes with displaced centers at Atla Regio is indicative of crustal adjustment above a long-lived mantle plume

Correspondence to:

P. J. Mason,
p.j.mason@imperial.ac.uk

Citation:

Mason, P. J., Klidas, A., Cirium, D., Ghail, R. C., & Lea-Wurzbach, S. (2025). Evolution of plume volcanism at Atla Regio, Venus. *Journal of Geophysical Research: Planets*, 130, e2024JE008815. <https://doi.org/10.1029/2024JE008815>

Received 4 NOV 2024

Accepted 3 FEB 2025

Author Contributions:

Conceptualization: P. J. Mason,

A. Klidas

Formal analysis: A. Klidas, D. Cirium, S. Lea-Wurzbach

Funding acquisition: P. J. Mason

Investigation: P. J. Mason, A. Klidas, D. Cirium, S. Lea-Wurzbach

Methodology: P. J. Mason, A. Klidas

Project administration: P. J. Mason

Supervision: P. J. Mason, R. C. Ghail

Visualization: A. Klidas

Writing – original draft: P. J. Mason, A. Klidas

Writing – review & editing: P. J. Mason, A. Klidas, R. C. Ghail

¹Department of Earth Science & Engineering, Imperial College London, London, UK, ²Department of Earth Atmospheric and Planetary Sciences, Purdue University, West Lafayette, IN, USA, ³Bullard Laboratories, Department of Earth Sciences, University of Cambridge, Cambridge, UK, ⁴Centre for Dynamic Earth and Solar System, Department of Earth Sciences, Royal Holloway, University of London, Egham, UK

Abstract Atla Regio is a large topographic rise, at the equator of Venus, considered to sit above a young mantle upwelling. Hosting several giant shield volcanoes, including Maat Mons, it is a strong candidate for recent eruptive activity. Through detailed analysis of material units and structures at Ozza, Maat and Sapas Mons, northern Dali Chasma and the surrounding plains, we have unraveled the tectono-magmatic evolution at Atla Regio. Lithostratigraphic analysis, of the volcanic styles, lava flow relationships, rift-associated shield clusters, graben-fissure systems and pit cratering, in relation to the regional plains and other older features, has enabled the establishment of a model of the evolution of volcanism. Structural analysis reveals that graben-fissure systems at Ozza Mons transition outward into three major rifts, and two new, unnamed volcanoes are identified. Our model is evidenced by analysis of lava flow stratigraphy and cross-cutting relationships with respect to the stratigraphic marker provided by the ejecta halo of the Uvaysi impact, and using sound stratigraphic principles. A relative chronostratigraphic framework for the area around Ozza and Maat Mons has thus been constructed, and our findings are in agreement with prior research. A new regional history is proposed here, where volcanism was initially concentrated along Dali Chasma, before plume volcanism took over at Ozza Mons and finally, Maat Mons became the most recent locus of volcanism. We propose that the pattern of large shield volcanoes with displaced centers here is indicative of minor crustal adjustments above a relatively long-lived mantle plume.

Plain Language Summary Atla Regio, near Venus's equator, lies above a rising section of the planet's mantle, which may be responsible for recent volcanic activity. This region is home to several large volcanoes, including Maat Mons, and shows signs of ongoing volcanic processes. By studying the volcanic features, lava flows, and geological structures around volcanoes like Ozza, Maat, and Sapas Mons, as well as the surrounding plains, a model to understand the evolution of volcanism in this area has been developed. This study examined how volcanic features, such as graben-fissure systems and pit craters, relate to older geological structures. The ejecta debris from the Uvaysi impact event has been used to date relatively in the region's volcanic history. The results show that volcanic activity in the area initially focused around Dali Chasma, then shifted to Ozza Mons, and later to Maat Mons, suggesting that a long-lived mantle plume may be driving the volcanic activity here. The spatial arrangement of these volcanoes, with their shifted positions, supports the idea of minor movements in the crust over time, allowing the mantle plume to influence volcanism in this region over a long period.

1. Introduction

1.1. Background

Despite being Earth's nearest neighbor, Venus may be the least well understood terrestrial planet in our solar system, but it is the one that potentially holds the most important clues about the evolution of such planets. With a hostile 720 K, 94 bar surface enshrouded by an optically impenetrable cloud deck, “Earth's twin” has eluded systematic scientific exploration that has been employed so successfully elsewhere; we have higher resolution imagery of the surface of Pluto than we do of Venus (77–80 m for Pluto from New Horizons and 120–300 m for Venus from Magellan). Hence, there are still many unanswered questions about Venusian geology. Almost everything we know comes from the near global imaging achieved by NASA's Magellan mission in 1990–1994, which revealed a complex landscape and abundant evidence for tectonic and volcanic activity. The origin and rates of these processes as well as their evolution through time are very poorly understood and fuel scientific investigation to this day.

© 2025. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

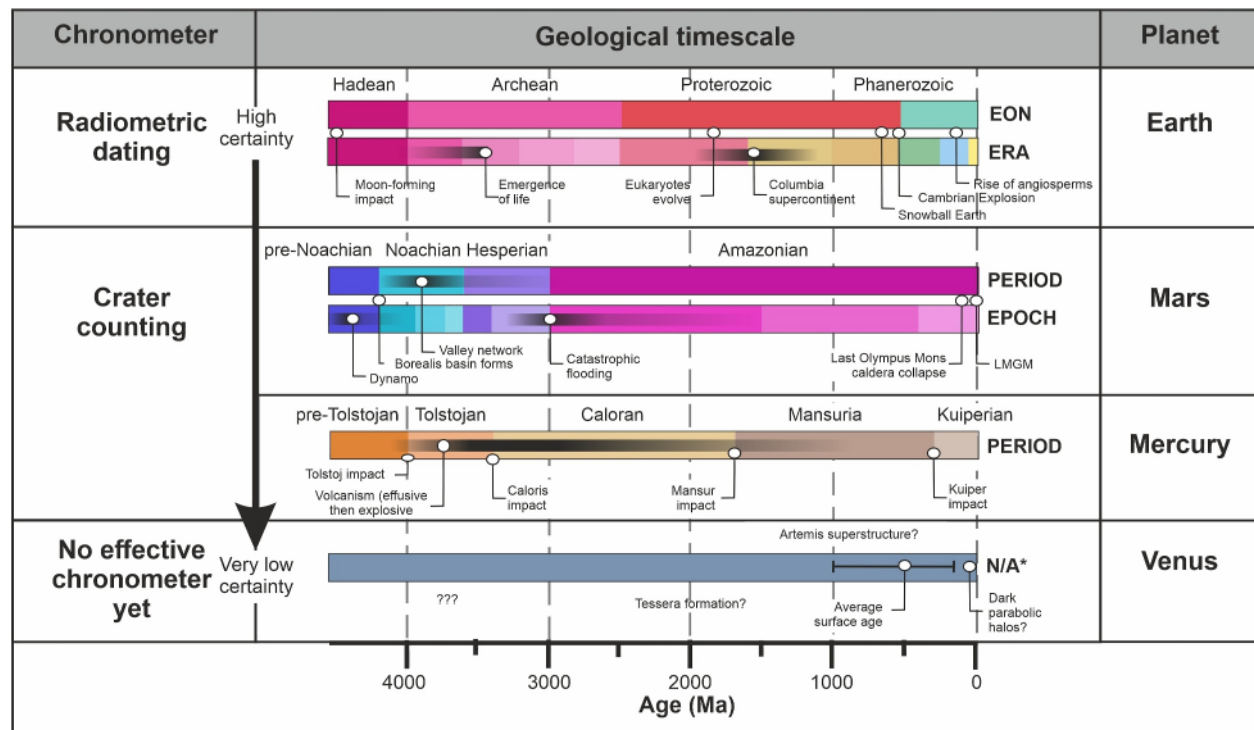


Figure 1. Approximate chronostratigraphic charts of the four terrestrial planets and key events. Earth time scale from International Chronostratigraphic Chart v2021/10. Mars time scale from Platz et al. (2013). Mercury time scale from Banks et al. (2017) and Thomas et al. (2014). * = A geological timescale for Venus was proposed by Ivanov and Head (2015) and was reviewed and updated by Head et al. (2023).

With low rates of crustal recycling, deposition and erosion, the Venusian surface preserves a rich record of volcanic and tectonic activity (Solomon et al., 1992). Inhibited by the planet's dry and viscous mantle (Nimmo & McKenzie, 1998), it is thought that terrestrial-style plate tectonics does not presently occur; strain is instead distributed across broad zones which are likely related to sites of mantle upwelling and downwelling (Solomon et al., 1991, 1992). Until recently, the Venusian lithosphere has been viewed as a relatively simple single plate, known as a "stagnant lid" (Solomatov & Moresi, 1996), though more complex models have been proposed in the last few years (Ghail, 2015; Tian et al., 2023). The planet's apparently random and uniform cratering distribution has been interpreted as evidence for a catastrophic global resurfacing event *ca* 500 Ma (Solomatov & Moresi, 1996; Strom et al., 1994; Turcotte et al., 1999), although the same cratering record is also consistent with steady-state equilibrium models of resurfacing (Bjornes et al., 2012). The catastrophic resurfacing model predicts that Venus should be geologically quiescent today, with insignificant rates of resurfacing in the recent geological past. Recent research challenges this idea, and provides evidence for significant and geologically recent (potentially ongoing) resurfacing, concentrated at rift zones (Brossier et al., 2022; Byrne et al., 2021; Gülcher et al., 2020; Shalygin et al., 2015) and volcanic rises (Brossier et al., 2021; Senske & Head, 1992; Shalygin et al., 2012; Smrekar et al., 2010) in historic times (Esposito et al., 1988; D'Incecco et al., 2021; Herrick & Hensley, 2023), in stark contrast to Mars, Mercury and the Moon, where most geological processes have ceased (Figure 1). This evidence suggests that we should expect many changes to be detected by the forthcoming new missions, both over the 40 years period since Magellan but also during the EnVision and VERITAS missions; much work now focuses on predicting the changes we might expect, and how they should best be detected (Campbell & Hensley, 2024; Gallardo i Peres et al., 2024; Hahn & Byrne, 2023).

1.2. The Nature of Volcanism on Venus

Venusian volcanism differs substantially in style from that on Earth or Mars. The high atmospheric pressure and inferred low volatile content of magmas suppress vesiculation and favor effusion, although some candidate pyroclastic deposits have been identified (Ganesh et al., 2021, 2022; Ghail & Wilson, 2015). In comparison to Earth, Venusian shields tend to be larger, because of the slow or non-existent rapid plate motion, have lower

slopes ($<1\text{--}3^\circ$), and lack evidence for circumferential faults. Commonly, they consist of a broad, almost flat lava apron surrounding a steep central edifice, and have abundant extensional radial structures. Volcanic edifice growth seems to involve a combination of extrusive and intrusive activity. Models suggest they are basally welded and supported by thick elastic lithospheres and large shallow magma chambers, as suggested by the caldera complexes often found at their summits (McGovern & Solomon, 1998; McGovern et al., 2014; Mougini-Mark, 2016; Stofan et al., 2001). As on Earth, fields of small volcanoes are also common on Venus. These consist predominantly of shields, though certain studies offer comparatively high counts of conical and domical volcanoes, sometimes associated with radar bright material which are inferred to be of pyroclastic origin (Guest et al., 1992; Kreslavsky & Head, 1999).

The classification of volcanoes on Venus largely follows that of Head et al. (1992) and other authors herein referenced; volcanoes are commonly divided, by size and abundance, into “vents” or “fields” and then sub-classified by morphology. This division reflects differences in magmatic budget; higher flux tends to produce large isolated vents, while volcanic fields form under restricted flux insufficient to build a single large volcano (Crumpler & Aubele, 2015). Such large features include large shield volcanoes (>100 km diameter), intermediate volcanoes (20–100 km) sub-classified into anemones, ticks and steep-sided domes and pateras, which are nominally defined as isolated calderas (Head et al., 1992). Small volcanoes, on the other hand, are some of the most abundant features on Venus and display strong clustering behavior; as such they have previously been mapped into fields and not at the vent level, until the work of Hahn and Byrne (2023). Such fields and their constituents have been studied in some detail by Addington (2001), Guest et al. (1992), Aubele (1993), Kreslavsky and Head (1999) and Ivanov and Head (2004).

1.3. Plume-Driven Tectonic and Volcanic Processes at Atla Regio

Recent magmatic activity on Venus is concentrated around the Beta-Atla-Themis (BAT) province, a geologically young area comprising interconnected Rift-Dominated Rises (RDRs) and regionally abundant tectono-magmatic features, such as large volcanoes, coronoids, shield clusters and flow fields (Airey et al., 2016; Guseva, 2016; Hansen & Young, 2007; Shalygin et al., 2012, 2015). RDRs on Venus are broad domal features associated with extension and large-scale volcanism. They are widely interpreted to be the surface expressions of mantle plumes, as their positive gravity anomalies and associated large apparent depths of isostatic compensation (~ 200 km) suggest active flexural support (Senske et al., 1992; Smrekar, 1994). The BAT province's RDRs are thought to represent multiple centers of triple-junction type rifting, with strong Earth analogs to plume-induced intraplate continental breakup (Ernst et al., 2007). Large triple junctions centered on Atla and Beta Regiones have been compared to the Afar system of East Africa (Figure 2) (Airey et al., 2016; Foster & Nimmo, 1996), and to other Large Igneous Provinces (Buchan & Ernst, 2021), while smaller incipient or failed triple junctions along Parga and Hecate Chasmata, located outside the radius of RDRs, have been compared to Atlantic rifting in the Mesozoic (Ernst & Desnoyers, 2004; Graff et al., 2018).

One such volcanic rise long thought to be a site of young volcanism and tectonism is Atla Regio (Basilevsky, 1993; Bilotti & Suppe, 1999; Brossier et al., 2021; Senske & Head, 1992; Shalygin et al., 2015), now confirmed by the recent discovery of visible surface changes to volcanic features at Maat Mons (Herrick & Hensley, 2023). Atla Regio has the highest swell height (~ 2.5 km) of any RDR and features the greatest volume of erupted volcanic material (Stofan et al., 1995) since it hosts four giant shield volcanoes. Oza Mons (OM) is the most voluminous of these and lies at the center of a gravity anomaly and at the tectonic junction of five rift zones (Senske et al., 1992), see (Figure 3). OM sits at the northern junction of Dali, Ganis, Parga and Hecate Chasmata. Maat Mons sits to the southwest of OM on the western flank of the Dali Chasma rift, and Sapas Mons lies some way off to the northwest of it. Oza Mons also lies at the western end of what is now proposed as the Great Dyke of Atla Regio (GDAR), a 3,700 km long single mafic dyke underlying Parga Chasma (El Bilali & Ernst, 2024), the continuity of which is taken as evidence of a temporal and kinematic link between OM and Parga Chasmata (Figure 4). Maat Mons (MM) is less broad but twice the height, and is the tallest volcano on Venus, reaching +9.6 km elevation. The third giant shield volcano, Ongwuti Mons, is poorly studied but appears highly deformed and associated with an equally giant coronoid feature. A fourth large volcano, Sapas Mons (SM), lies some 1,000 km to the NW of the main Atla swell; this seems to represent an off-axis site of upwelling in its own right (Schinella et al., 2011) and its relationship to Atla is unclear. Atla Regio has been mapped at 1:10M as part of the Ivanov and Head (2015) global geological map, though detailed mapping by the USGS at 1:5M has not been conducted in quadrangles V-26 or V-38, which cover the Atla Regio area.

3.3.3. “Aeronwen Mons”

An additional unnamed intermediate-sized shield volcano with a relief of ~ 2.75 km lies along Dali Chasma roughly midway between Ongwuti and Ozza Montes, located at 161.4°W 1.8°N (Figure 4). Here named “Aeronwen Mons” (AM), pending IAU approval, it is heavily deformed by DC structures to the point where its lava apron is no longer preserved or visible. El Bilali et al. (2023) found a radial lineament network interpreted as a radial dyke swarm associated with this feature, which is further evidence that this construct is volcanic in origin. On its western flank, a partially overprinted caldera structure is preserved, with a ~ 20 km diameter elliptical depression post-dated by at least three subsequent smaller circular collapse structures. Since the Uvaysi impact crater is superimposed on top of the AM flank, volcanic activity was likely restricted to the lower Atla era.

3.3.4. Ozza Mons

OM is a broad, low volcano with a $1,000 \times 800$ km apron, but while the elevation of its summit reaches 7.5 km, the relief of the volcano is only ~ 3.5 km, and its flanks are very gentle sloping. The upper slopes are dominated by a homogenous radar-bright (Li) unit, and the lower flanks by extensive radar-dark (Ls) flows which embay the (Li) unit, reaching ca 750 km from the summit. The flanks are heavily deformed by a radial graben-fissure system and by three incipient rifts. Many lava flows are seen to infill the graben and overprint rift structures, but the reverse is also common, suggesting that at OM rifting and volcanism were coeval.

The literature is divided on whether Ozza Mons hosts a “tectonized caldera” (Brossier et al., 2021) or a 1.5-km high radar-dark oval plateau (Senske & Head, 1992) at its summit and it is unlikely that both are true. The Magellan GDTR product shows a prominent elevated plateau, but the patterns of radar-brightness and -darkness in the SAR imagery support a caldera hypothesis. The extremely low radar return of the 80 km diameter oval-shaped region is expected from the smooth texture of ponded lava. Both the floor and rim of the caldera host small domes and pit craters, some of which are very large (< 20 km in their longest axis). The margins host circumferential structures which may be normal faults. As the rim appears degraded and partially buried by a younger SC lava field, the larger structure is interpreted to be a relict caldera, perhaps deformed by down-sagging. If this interpretation is correct, then OM may host the largest shield volcano summit caldera on Venus yet identified. Such circumferential structures are also seen at Sif Mons (Stofan et al., 2001), Zeus patera on Olympus Mons (Mouginis-Mark, 2021), and Arsia Mons. The caldera hypothesis may be supported by the fact that this structure lies at the convergence point of a series of pit crater chains, which suggests the presence of a magma chamber.

The effects of weathering have rendered the earliest exposed lava flows of Ozza Mons (OM1) difficult to separate; they are so radar bright and featureless that the individual flows are indistinguishable. These were followed by more episodes of high volume, likely dyke-fed eruptions, further from the summit, forming the OM2 and OM4 sheet flows. These are extensive and may have significantly drained the magma chamber and triggered caldera collapse(s). Near-summit units such as OM6 and the uppermost OM3 flows may have originated either from caldera overflow events or from circumferential faults forming ring dykes. The OM7 unit is interpreted to represent a lava lake that formed and ponded during the last large caldera collapse event at OM.

Following this, the most recent phase of volcanic activity formed the cluster of shields (OM8), which partially filled the relict caldera and incipient Ganis Chasma graben. The summit caldera is no longer quasi circular but is significantly deformed and now elongated in form; it may be relatively old in comparison to the morphologically fresh summit caldera at Maat Mons. This suggests that the magma chamber either remains partially molten but has not been replenished recently or it has cooled and crystallized. In sharp contrast to MM, OM's flank lavas show low spatial variability of emissivity, and rifting has propagated far up toward the summit. These features hint that Ozza Mons is older, and thus a cooled and inactive magma chamber may be favored. Stratigraphic analysis shows that the bulk of the volcanic construct pre-dates the Uvaysi impact PRDH and formed during the lower Atla era; however, since the PRDH does not extend to the summit region, it cannot be ruled out that the youngest units OM7-8 were emplaced in the Atla era's upper phase, contemporaneously with MM. The nature of the youngest volcanic deposit, the small OM8 summit shield cluster, indicates that overall volcanism at Ozza Mons has waned over time. However, sporadic low volume eruptions in this shield cluster could conceivably continue to the present day. On longer timescales, the magma chamber could 1 day be replenished and the volcano could re-enter a more energetic phase.

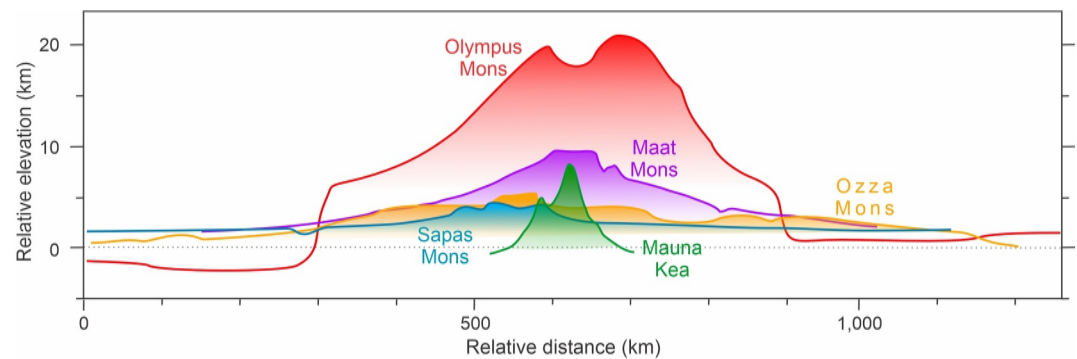


Figure 7. Relative topographic profiles across Maat, Ozaa and Sapas Montes, in comparison to those of Olympus Mons, Mars and Hawaii, Earth, contrasting the size and extent of these volcanoes. Elevation sources: Global Topographic Data Record (GTDR-SINUS.3;2), Venus, Mars Orbital Laser Altimeter (MOLA) and Earth Shuttle Radar Topography Mission (SRTM V3, 1 arc-second).

3.4. Later Volcanic Activity (Atla Era, Upper Phase)

The upper Atla era begins at the time of the Uvaysi crater (UC) impact and the deposition of its Parabolic Radar Dark Halo (PRDH) apron, which forms a widespread stratigraphic marker readily distinguishable in radar imagery that has been tentatively dated at less than 60–9 Ma (Brossier et al., 2021). The Uvaysi event coincides with a significant shift in the center of active volcanism at the Atla Regio. At this juncture, volcanism along Dali Chasma seems to have ceased, although tectonism clearly continued, since there are fractures which cross-cut the Uvaysi crater rim and ejecta. At this point, Ozaa Mons may be in decline or dormant while Maat Mons becomes the main site of active volcanism. UC lies in Dali Chasma on the northern flank of Aeronwen Mons and has a ~35 km diameter and a radar-bright floor. The UC airfall ejecta PRDH blankets all the older OM units but is post-dated by MM lava flows (Figures 6 and 9).

It seems clear that volcanic activity at Ongwuti, Aeronwen, and Ozaa Montes has declined or terminated at this time, and the same may be true for tectonism at Dali Chasma (as no DC structures overprint the MM3 flows that have flooded it), though the reason for the decline is unclear. Perhaps successive eruptions at the three older volcanoes built the lithosphere up to a point where magma ascent is more difficult, and so the replenishment of magma chambers eventually declined or ceased. This would cause migration of melt to a new site with thinner lithosphere, where MM is located now, but the fact that MM is twice the height of OM challenges that idea.

The unusual profile of Maat Mons is another mystery, being much steeper than Ozaa or Sapas Montes (Figure 7); could this be a result of differing lava composition? Perhaps the lavas of the lower Atla era represent early higher temperature, low-viscosity “plume-head” melts, whereas the upper Atla era MM lavas were sourced from lower temperature melts that were more viscous, flowed less far, accumulated faster and constructed a steeper edifice. Either Maat formed recently in the Upper Atla Era or it was present in the lower Atla era but resurfaced its flanks later in the Upper Atla Era. The lower diameter and structural complexity of Maat’s summit caldera and the high frequency of radar bright flows, as compared to Ozaa’s, suggest that it is significantly less mature.

3.4.1. Maat Mons

Maat Mons is the tallest volcano on Venus, and its summit is twice as high as that of OM despite having a less broad apron [Figure 8]. As at OM, the flanks at low elevation are dominated by extensive Ls flows which flood local topography, but at high elevations there is no equivalent to the radar-bright homogenous (Li) unit. Instead, discrete (Lf and Ld) flows are present, varying in radar brightness and either emanating from the summit caldera region or from lower flank vents. Maat Mons’ high radar-emissivity summit is anomalous compared to other high elevation regions on Venus, such as Ozaa, Sapas, and Maxwell Montes, which have a very low radar-emissivity. This has been interpreted to indicate a relatively young age of emplacement (Brossier et al., 2021; Klose et al., 1992), although a low radar-emissivity (Li) region on the SW flank may be older. It is noteworthy that no lineaments or structures are found to overprint MM’s lava flows, despite the fact that distal (Ls) flows flood large areas of Dali Chasma’s rift (Figure 5). This is in strong contrast to OM, which has been partially overprinted by rift

Mons. Two newly described intermediate-sized shield volcanoes have also been identified (here called Tara and Aeronwen Montes), which shed light on the region prior to and just after plume arrival. Stratigraphic analysis of Ozza Mons reveals eruptions shifting from the western to eastern flanks and becoming increasingly restricted to the summit region through time. Its flanks are also deformed by radial graben-fissures and three rift systems approach the summit region. The radar-dark region at Ozza Mons' summit represents a deformed, relict caldera, potentially the largest summit caldera yet identified on Venus. Increasingly centralized eruptions of reducing extent at Ozza Mons suggest that volcanism along this axis is waning, although there is currently insufficient evidence to determine if the volcano is extinct or simply in a phase of inactivity.

The Uvaysi impact and its parabola of ejecta material form a very useful stratigraphic marker, being clearly distinguishable as post-dating lava flows on the SW flank of OM and pre-dating flows on the NW flank of MM. Having been tentatively dated between 9 and 60 Ma, this impact marks an important boundary between Lower and Upper Atla Era development. Using this local marker, in addition to the global marker of regional plains (corresponding to time T), and sound principles of superposition in mapping, a chronostratigraphic framework for Atla Regio is proposed here. This framework consists of several key elements: 1. "Pre-Atla era" consisting of tessera, plains emplacement, early shield volcanism and the newly described early "Tara Mons"; 2. "Lower Atla era," where early plume-induced magma-assisted rifting at Dali Chasma produces three large shield volcanoes: Ongwuti Mons, Ozza Mons, and the newly described "Aeronwen Mons." Evidence suggests that volcanism at OM is declining before the Uvaysi impact event; and 2. "Upper Atla era," where MM becomes the locus of active volcanism in Atla Regio, and which lasts until the present day; as supported by the recently discovered expansion of a caldera on the northern flank of Maat Mons (Herrick & Hensley, 2023) and in agreement with El Bilali et al. (2023) and El Bilali and Ernst (2024). This framework is in good correlation with the revised global framework proposed by Head et al. (2023). The younger Maat Mons flows, postdate the Uvaysi impact event and flood areas of Dali Chasma rift zone and yet are notably undeformed. Maat Mons also possesses a smaller and structurally simpler summit caldera and lacks a radial graben-fissure system. All these features suggest that Maat Mons is less mature than Ozza Mons and that it is active today.

5. Conclusions

We propose here a picture of active but shifting volcanism, caused by minor crustal shifts and adjustments, above a long-lived and static mantle plume system, likely driven by active rifting processes to the east, across the BAT province. With a second plume likely located beneath Themis Regio, which may be more or less active than the Atla plume at any one time, differential plume activity and accompanying volcanism may drive further crustal jostling behavior here. Our detailed mapping demonstrates that volcanism at Atla developed at several different centers over time. The identification of the Uvaysi impact PRDH as post-dating lavas at Ozza Mons but predating lavas at Maat Mons has provided a vital stratigraphic marker and has enabled the construction of a relative stratigraphic framework for this region. In time, if Maat remains the center of active volcanism for a considerable period of time, a new triple-junction style rift system may develop here, and some of the existing "arms" of the rift may decay and revert to what are classified elsewhere as "groove belts"; only time will tell. Nevertheless, the BAT region is clearly a site where significant change and evidence of young active volcanism should be expected in the data collected by the next surface imaging missions to Venus, EnVision and VERITAS.

Data Availability Statement

The primary data used was acquired by the Magellan mission. These data included 75 m Synthetic Aperture Radar (SAR) imagery (https://astrogeology.usgs.gov/search/map/Venus/Magellan/Venus_Magellan_LeftLook_mosaic_global_75m.xml) and the GTDR global topographic data product, GTDR-SINUS.3;2 (Ford, 1992a, 1992b; Morgan, 1994, https://astrogeology.usgs.gov/search/map/Venus/Magellan/RadarProperties/Venus_Magellan_Topography_Global_4641m_v02). Other elevation data sources include the Mars Orbital Laser Altimeter (MOLA, https://astrogeology.usgs.gov/search/map/mars_mgs_mola_dem_463m) and Earth Shuttle Radar Topography Mission (SRTM V3, 1 arc-second, <https://www.earthdata.nasa.gov/data/instruments/srtm>). The global geological terrain units, as published by Ivanov and Head (2011), were used and simplified to create the regional geological map shown here.